

Kyungmin Kim

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🎓 [Google Scholar Profile](#)

🌐 [kyungmin-git.github.io](https://github.com/kyungmin-git)

Education

- 2021 – 2026 • **Ph.D. in Geophysics, University of Alaska Fairbanks (UAF)**
Defense date: September 8, 2026
Advisor: Dr. Tàrsilo Girona
Committee: Dr. Carl Tape, Dr. Kyle Anderson, Dr. Matt Haney
Relevant Coursework: Foundations of Geophysics, Presentation Techniques in Geosciences, Applied Multivariate Statistics, Inverse Problems and Parameter Estimation, Selected Topics in Volcanology, Nonparametric Statistics, Numerical Linear Algebra, Numerical Analysis of Differential Equations, Applied Seismology
- 2018 – 2020 • **M.S. in Earth Science, Tohoku University**
Scholarship: Japanese Government (MEXT) Scholarship
Advisor: Dr. Michihiko Nakamura
- 2017 • **B.S. in Earth and Environmental Sciences, Seoul National University**
Advisor: Dr. Jinho Ahn

Research Experience

- 2021 – 2026 • **UAF Geophysical Institute (GI), Research Assistant**
Fairbanks, AK
Identifying Long-Period (LP) seismicity from the Great Sitkin Volcano and developing a Bayesian inversion technique to invert its subsurface physical properties.
Applying this technique with data assimilation of seismicity and tremor to track the volcano's eruptive activity since 2021.
- 2023 • **Istituto Nazionale di Geofisica e Vulcanologia (INGV), Visiting Student(Sep – Dec 2023)**
Rome, Italy
Conducted laboratory experiments simulating gas-driven volcanic seismicity, injecting gas through conduits filled with water and porous media to replicate subsurface flow conditions.
- 2018 – 2020 • **Tohoku University, Graduate Student Researcher**
Sendai, Japan
Performing petrological high-temperature experiments to investigate the eruption temperature and pressure conditions of the Hoei eruption of Fuji Volcano (1707 CE).
Investigating 1-D conduit flow dynamics of Basaltic Plinian eruptions and exploring the conduit flow dynamics of the Hoei eruption of Fuji Volcano based on petrological/geological constraints from experiments and geological surveys.
- 2015 – 2017 • **Seoul National University, Undergraduate Student Researcher**
Seoul, South Korea
Reconstructed paleoclimate conditions and characterized the geochemical environment of permafrost in central Yakutia by analyzing trapped gas compositions from ice-wedge samples.

Publications

- 2023 • Yang, J. W., Ahn, J., Iwahana, G., Ko, N., Kim, J. H., **Kim, K.**, et al., & Han, S. (2023). Origin of CO₂, CH₄, and N₂O trapped in ice wedges in central Yakutia and their relationship. *Permafrost and Periglacial Processes*, 34(1), 122–141. [doi:10.1002/ppp.2176](https://doi.org/10.1002/ppp.2176)
- 2020 • Yang, J. W., Ahn, J., Iwahana, G., Han, S., **Kim, K.**, Fedorov, A. (2020). Brief Communication: The reliability of gas extraction techniques for analysing CH₄ and N₂O compositions in gas trapped in permafrost ice wedges. *The Cryosphere*, 14(4), 1311–1324. [doi:10.5194/tc-14-1311-2020](https://doi.org/10.5194/tc-14-1311-2020)
- 2019 • **Kim, K.**, Yang, J. W., Yoon, H., Byun, E., Fedorov, A., Ryu, Y., Ahn, J. (2019). Greenhouse gas formation in ice wedge at Cyuie, central Yakutia. *Permafrost and Periglacial Processes*, 30(1), 48–57. [doi:10.1002/ppp.1994](https://doi.org/10.1002/ppp.1994)

Paper Under Review

- 2026 • **Kim, K.**, Girona, T., Spina, L., Taddeucci, J., Taryn, L., Clor, L., Pennacchia, F., Cornelio, C., Spagnuolo, E. Experimental insights into gas flow-driven seismicity in porous media: Estimating volcanic gas emissions from seismic data. Submitted to *Geophysical Research Letters* (MS ID: 2026GL126333).
- **Kim, K.**, Girona, T., Anderson, K., Haney, M. Characterizing the onset of the 2021 Great Sitkin dome-building eruption through the trans-dimensional Bayesian inversion of long-period seismicity. Submitted to *JGR: Solid Earth* (MS ID: 2026JB035613).

Paper Under Preparation

- **Kim, K.**, Girona, T., Power, J. Ensemble data assimilation of long-period seismicity during the Great Sitkin eruption since 2021: Implications for gas flow rate and eruption dynamics.
- Sohn, R., **Kim, K.**, Girona, T., Caudron, C., Tremor and explosive potential of steam pockets beneath Yellowstone Lake.

Work Experience

Mar 2013 – Feb 2015 • **Weather Observer**, Seoul Air Base

Seongnam, South Korea

Completed mandatory military service (Republic of Korea) as a Sergeant. Observed and recorded weather conditions on an hourly basis and prepared data for daily weather briefings.

Awards

- Mar 2018 – Feb 2020 • **MEXT Scholarship**, Japanese Ministry of Education, Culture, Sports, Science and Technology
Provided a full stipend and tuition coverage for Master's study in Japan.
- 2024 • **Travel Grant**, University of Alaska Fairbanks (UAF)
Awarded conference travel funding to attend the Seismological Society of America Annual Meeting (Anchorage, AK).

Conference Presentations

- 06/2025
 - IAVCEI Scientific Assembly, Geneva, Switzerland. Poster Presentation. “Characterizing pre- and syn-eruptive processes at Great Sitkin Volcano, Alaska, by integrating seismic data assimilation and satellite-based thermal anomalies.” Kim, K., Girona, T., Anderson, K., Power, J., Yang, X., Haney, M.
 - IAVCEI Scientific Assembly, Geneva, Switzerland. Poster Presentation. “Investigating volcanic tremor and long-period seismic events through experiments on porous media gas flow.” Kim, K., Spina, L., Taddeucci, J., Pennacchia, F., Cornelio, C., Spagnuolo, E., Girona, T.
- 12/2024
 - American Geophysical Union Annual Meeting, Washington, DC. Oral Presentation. “Characterization of the 2021–2022 Great Sitkin dome-building eruption through Bayesian inversion of LP seismicity.” Kim, K., Girona, T., Anderson, K.
- 09/2024
 - NASA Solid Earth Team Meeting, Washington, DC. Poster Communication. “Constraining the pre-eruptive and co-eruptive conditions of dome-building eruptions through the analysis of thermal anomalies and the inversion of long-period seismicity: the case of Great Sitkin Volcano, Alaska.”
- 04/2024
 - Seismological Society of America Annual Meeting, Anchorage, AK. Poster Communication. “Laboratory experiments on gas-driven volcanic tremor and long-period seismicity.” Kim, K., Spina, L., Taddeucci, J., Pennacchia, F., Cornelio, C., Spagnuolo, E., Girona, T.
 - Seismological Society of America Annual Meeting, Anchorage, AK. Poster Communication. “Characterization of the onset of the Great Sitkin dome-building eruption through the trans-dimensional Bayesian inversion of LP seismicity.” Kim, K., Girona, T., Anderson, K.
- 02/2023
 - IAVCEI Scientific Assembly, Rotorua, New Zealand. Oral Presentation (Virtual). “Characterization of the 2021–2022 Great Sitkin dome-building eruption through Bayesian inversion of LP seismicity.” Kim, K., Girona, T., Anderson, K.
 - IAVCEI Scientific Assembly, Rotorua, New Zealand. Poster Communication (Virtual). “Petrological constraints on the magma fragmentation pressure of the 1707 Hiei eruption of Fuji Volcano and inferred eruption dynamics.” Kim, K., Nakamura, M., Kozono, T., Yoshimoto, M.
- 06/2022
 - Cities on Volcanoes, Heraklion, Greece. Oral Presentation (Virtual). “Characterization of the 2021–2022 Great Sitkin dome-building eruption through the inversion of LP seismicity.” Kim, K., Girona, T.
- 09/2019
 - Volcanological Society of Japan, Kobe, Japan. Oral Presentation. “Petrological constraints on the magma fragmentation pressure on the 1707 CE eruption (Hiei) of Fuji Volcano – implications for the dynamics of basaltic Plinian eruptions.” Kim, K., Nakamura, M., Kozono, T., Yoshimoto, M.
- 06/2016
 - International Conference on Permafrost, Potsdam, Germany. Poster Communication. “Ancient environments and climate of central Siberia inferred from ice wedge at Cuyie Village near Yakutsk.” Kim, K., Ahn, J., Byun, E., Yoon, H., Fedorov, A., Yang, J. W.

Field Work Experience

- 06/2025
 - IAVCEI SA2025 field trip to Eifel Volcanic Field, Germany
- 08/2024
 - Alaska Volcano Observatory field work to Redoubt, Iliamna, and Ninilchik receiver sites (with Ellie Boyce); helped fix network sites and receivers
- 05/2018
 - Asama Volcanic Field excursion

Field Work Experience (continued)

07/2015 • Field work to Yakutsk and Cyuie Village, central Yakutia, Russia; collected ice-wedge samples at field sites

Student Activity

May 2022 – Aug 2023 • **Treasurer, Geophysical Institute Graduate Student Association (GIGSA)**

Fairbanks, AK

Managed the association's accounts and finances.

Languages

Korean • Native language (spoken, written)

English • Advanced (spoken, written)

Japanese • Advanced (spoken, written)